

Keil code:

```
#include <reg51.h>
```

```
sbit LED0 = P1^0;
```

```
sbit LED1 = P1^1;
```

```
void delay()
```

```
{
```

```
    unsigned int i, j;
```

```
    for(i = 0; i < 500; i++)
```

```
    {
```

```
        for(j = 0; j < 100; j++);
```

```
    }
```

```
}
```

```
void main()
```

```
{
```

```
    while(1)
```

```
    {
```

```
        P1 = 0xFF;
```

```
        delay();
```

```
        P1 = 0x00;
```

```
        delay();
```

```
        LED0 = 1;
```

```
        LED1 = 0;
```

```
        delay();
```

```
        LED0 = 0;
```

```
        LED1 = 1;
```

```
        delay();
```

```
    }
```

```
}
```

— this code use if the above one is not working —

```
#include <reg51.h>
```

```
sbit LED0 = P1^0;
```

```
sbit LED1 = P1^1;
```

```
sbit LED2 = P1^2;
```

```
sbit LED3 = P1^3;
```

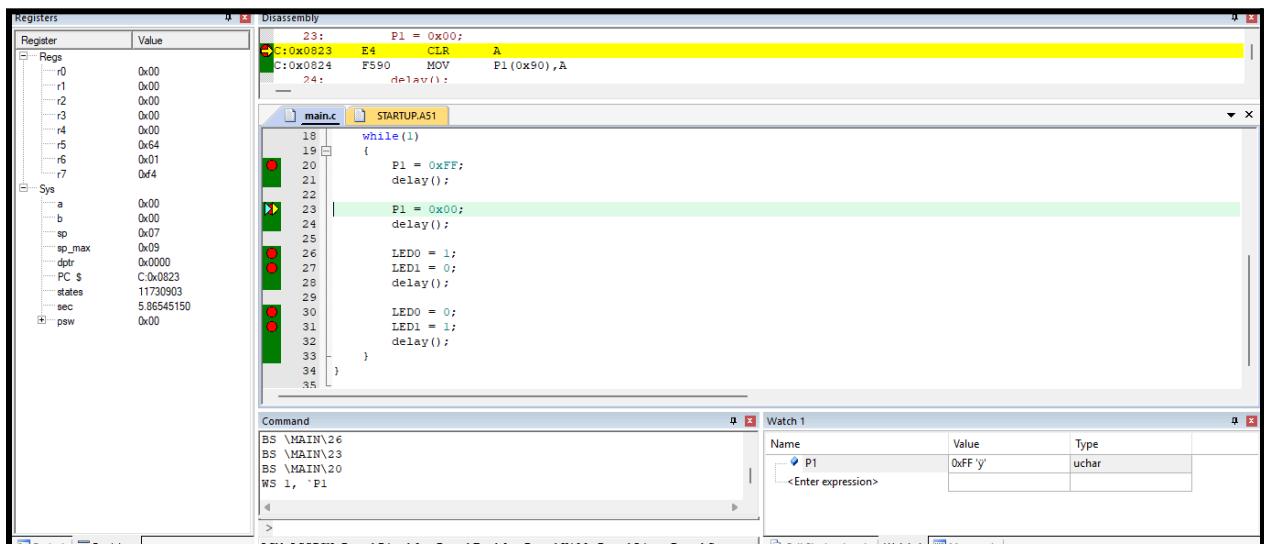
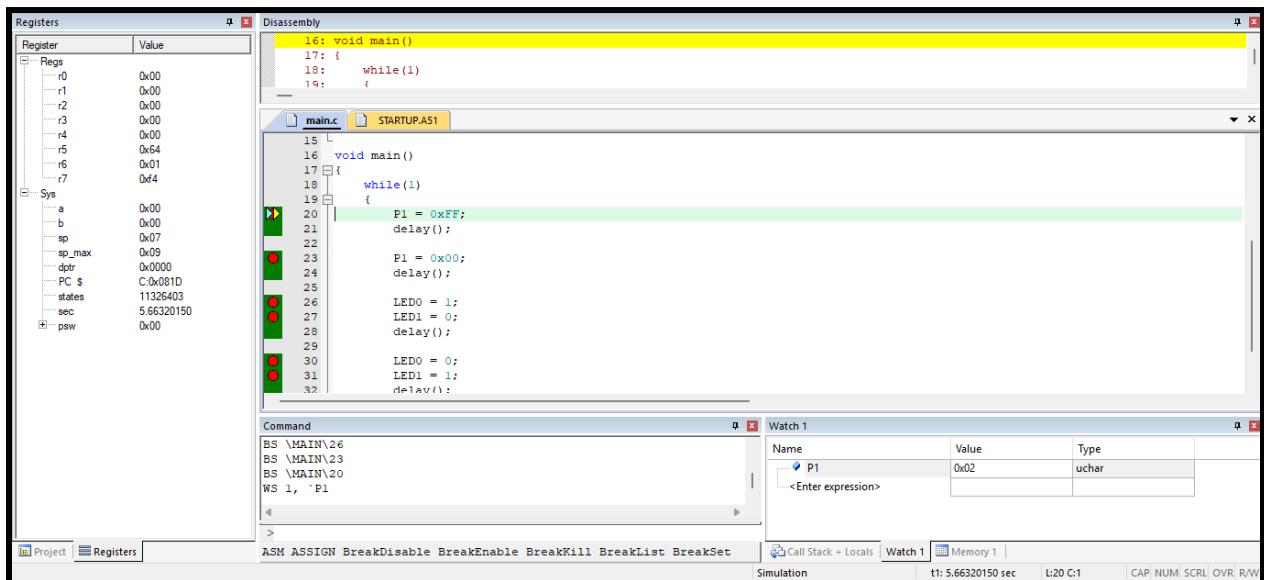
```

void main()
{
    P1 = 0x00;

    LED0 = 1;
    LED1 = 1;
    LED2 = 1;
    LED3 = 0;

    while(1)
    {
    }
}

```



Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x64
r6	0x01
r7	0x44
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x09
dptr	0x0000
PC	0x0829
states	12135403
sec	6.06770150
psw	0x00

Disassembly

```

26:      LED0 = 1;
C:0x0829 D290 SETB LED0(0x90.0)
27:      LED1 = 0;
C:0x082B C291 CLR LED1(0x90.1)
28:      delay();
29:
30:      LED0 = 0;
31:      LED1 = 1;
32:      delay();
33:  }
34:
35:

```

main.c STARTUP.A51

```

19:  {
20:      P1 = 0xFF;
21:      delay();
22:
23:      P1 = 0x00;
24:      delay();
25:
26:      LED0 = 1;
27:      LED1 = 0;
28:      delay();
29:
30:      LED0 = 0;
31:      LED1 = 1;
32:      delay();
33:  }
34:
35:

```

Command

```

BS \MAIN\26
BS \MAIN\23
BS \MAIN\20
WS 1, 'P1

```

Watch 1

Name	Value	Type
P1	0x00	uchar
<Enter expression>		

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x64
r6	0x01
r7	0x44
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x09
dptr	0x0000
PC	0x082B
states	12135404
sec	6.06770200
psw	0x00

Disassembly

```

27:      LED1 = 0;
C:0x082B C291 CLR LED1(0x90.1)
28:      delay();
29:
30:      LED0 = 0;
31:      LED1 = 1;
32:      delay();
33:  }
34:
35:

```

main.c STARTUP.A51

```

19:  {
20:      P1 = 0xFF;
21:      delay();
22:
23:      P1 = 0x00;
24:      delay();
25:
26:      LED0 = 1;
27:      LED1 = 0;
28:      delay();
29:
30:      LED0 = 0;
31:      LED1 = 1;
32:      delay();
33:  }
34:
35:

```

Command

```

BS \MAIN\26
BS \MAIN\23
BS \MAIN\20
WS 1, 'P1

```

Watch 1

Name	Value	Type
P1	0x01	uchar
<Enter expression>		

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x64
r6	0x01
r7	0x44
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x09
dptr	0x0000
PC	0x0830
states	12539903
sec	6.26995150
psw	0x00

Disassembly

```

30:      LED0 = 0;
C:0x0830 C290 CLR LED0(0x90.0)
31:      LED1 = 1;
C:0x0832 D291 SETB LED1(0x90.1)
32:      delay();
33:  }
34:
35:

```

main.c STARTUP.A51

```

19:  {
20:      P1 = 0xFF;
21:      delay();
22:
23:      P1 = 0x00;
24:      delay();
25:
26:      LED0 = 1;
27:      LED1 = 0;
28:      delay();
29:
30:      LED0 = 0;
31:      LED1 = 1;
32:      delay();
33:  }
34:
35:

```

Command

```

BS \MAIN\26
BS \MAIN\23
BS \MAIN\20
WS 1, 'P1

```

Watch 1

Name	Value	Type
P1	0x01	uchar
<Enter expression>		

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x64
r6	0x01
r7	0x64
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x09
dptr	0x0000
PC \$	C:0x0832
status	12539904
sec	6.26995200
psw	0x00

Disassembly

31: LED1 = 1;

C:0x0832 D291 SETB LED1(0x90.1)

32: delay();

C:0x0834 120R00 T.CAT.L delay(C:0800)

main.c

STARTUP.A51

19 {

20 P1 = 0xFF;

21 delay();

22

23 P1 = 0x00;

24 delay();

25

26 LED0 = 1;

27 LED1 = 0;

28 delay();

29

30 LED0 = 0;

31 LED1 = 1;

32 delay();

33 }

34 }

35 }

Command

BS \MAIN\26

BS \MAIN\23

BS \MAIN\20

WS 1, 'F1

Watch 1

Name	Value	Type
P1	0x00	uchar
<Enter expression>		